



Brief paper

Synchronizing nonlinear complex networks via switching disconnected topology[☆]Yao Chen^a, Wenwu Yu^{b,1}, Shaolin Tan^c, Henghui Zhu^d^a Department of Computing, The Hong Kong Polytechnic University, Hong Kong^b Department of Mathematics, Southeast University, Nanjing 210096, China^c College of Electrical and Information Engineering, Hunan University, Changsha 410082, China^d Division of Systems Engineering, Boston University, Boston, MA 02215, USA

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ABSTRACT

The current theoretical study on the synchronization mechanism of complex networks mainly focuses on fixed connected topology or switching balanced topology, which is heavily based on constructing common smooth Lyapunov functions. However, for complex networks with nonlinear node dynamics, it is still unknown whether such kind of networks can synchronize under the condition of switching, directed, and disconnected topology. By employing the ideas of sequential connectivity and joint connectivity, this paper finds that complex networks with one-sided Lipschitz node dynamics can realize synchronization even if the network topology is not connected at any time instant. By iteratively estimating the maximal distance between different nodes, this paper gives several sufficient conditions on synchronization of nonlinear complex networks under switching disconnected topology. Finally, simulation examples validate the main results.

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1. Introduction

Complex networks are a typical type of complex systems, representing the complex interactions among different components. Multitudes of social, biological and engineering systems can be described and analyzed by complex networks (Albert & Barabási, 2002). Historically, the investigation of complex networks can be traced back to the discovery of the fundamental scale-free and small-world networks (Strogatz, 2001; Watts & Strogatz, 1998). Based on these innovative models, many interesting results have been reported in the field of complex networks, towards to the understanding of the origin of complexity.

Synchronization is one of the typical collective behaviors of complex networks (Arenas, Diaz-Guilera, Kurth, Moreno, & Zhou,

2008), which means the states of a system reach some identical value or trajectory asymptotically. Synchronization also has some important engineering applications, such as clock synchronization of sensor networks and distributed power generation (Blaabjerg, Teodorescu, Liserre, & Timbus, 2006). During the past decade, a great number of interesting results have been reported in the field of network synchronization from different perspectives (Lü & Chen, 2005). It should be pointed out that most of the existing results only concern the dynamical behavior of complex networks with fixed topology, which can be generally attributed to the master equation method (Li, Duan, Chen, & Huang, 2010; Pecora & Carroll, 1998) or Lyapunov method (Yu, Chen, & Lü, 2009). Nevertheless, due to the complex environment in which the networks are involved, the network topology itself may also be dynamic, called dynamic or switching topology. However, there are few reported results focusing on synchronization of nonlinear complex networks with switching topology, especially disconnected topology.

The reliability of communication links in complex networks plays a critical role for network synchronization. Though existing results demonstrate the synchronizability of complex networks under the condition of fixed topology (Li et al., 2010), it remains an interesting issue to ask whether complex networks can realize synchronization under imperfect communication. Recently, in the field of multi-agent systems (MAS), some interesting results

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E-mail addresses: chenyao07@gmail.com (Y. Chen), wwyu@seu.edu.cn (W. Yu), shaolintan@hnu.edu.cn (S. Tan), henghuiz@bu.edu (H. Zhu).

¹ Corresponding author (W. Yu). Tel.: +86 25 52090597; fax: +86 25 52090590.

demonstrate that MAS can reach synchronization under the condition of fast switching topology (Kim, Shim, Back, & Seo, 2013), jointly connected and sequentially connected topology (Angeli & Bliman, 2009; Chen, Lü, Yu, & Hill, 2013; Chen, Yu, Li, & Feng, 2013; Hong, Gao, Cheng, & Hu, 2007; Jadbabaie, Lin, & Morse, 2003; Ren & Beard, 2005). Here, joint (or sequential) connectivity means the topology is not necessarily connected at any time instant but connected over a time interval (Cao, Morse, & Anderson, 2008). Inspired by these findings, it is natural to ask whether complex networks can reach synchronization under the condition of sequential or joint connectivity.

As far as we know, there does not exist a result which discusses the synchronization of nonlinear complex networks under the condition of jointly connected topology. Furthermore, even for the simplest case of networked linear system with switching topology, the widely used quadratic or smooth Lyapunov function does not exist (Brayton & Tong, 1979; Olshevsky & Tsitsiklis, 2008). Considering the nonlinear dynamics involved in the model of complex networks, solving such a problem becomes even more difficult. Therefore, it is necessary to develop novel techniques to handle the synchronization problem in case of switching topology. The major contribution of this paper is: it employs the ideas of sequential and joint connectivity, and proves that complex networks with one-sided Lipschitz node dynamics can realize synchronization under the condition of switching, directed, and disconnected topology.

This paper is organized as follows: Section 2 gives some preliminaries on graph theory; Section 3 presents the main results; Section 4 illustrates the details of technical proof; Section 5 gives a simulation example to verify the main results; Section 6 concludes this paper.

2. Preliminaries

A graph $\mathcal{G} = \{V, E\}$ is composed of two sets: V is the set of nodes and $E \subseteq V \times V$ is the set of edges. Given two graphs $\mathcal{G}_1 = \{V, E_1\}$ and $\mathcal{G}_2 = \{V, E_2\}$, the union of these two graphs is defined by $\mathcal{G}_1 \cup \mathcal{G}_2 = \{V, E_1 \cup E_2\}$. A graph $\mathcal{G} = \{V, E\}$ is called strongly connected if there exists a path from any node $i \in V$ to $j \in V$ ($i \neq j$). A graph $\mathcal{G} = \{V, E\}$ is called connected if for any pair of vertices i, j ($i \neq j$), there exists a path from i to j or from j to i . If $\mathcal{G}$ is connected, then $\mathcal{G}$ contains a spanning tree. In this paper, we focus on complex networks with topologies described by directed graphs.

Given any nonnegative matrix $A = (a_{ij})_{i,j=1}^N \in \mathbb{R}^{N \times N}$, the graph corresponding to A is given by $\mathcal{G}(A) = \{V, E\}$, that is $(j, i) \in E$ if and only if $a_{ij} > 0$.

Given a graph $\mathcal{G} = \{V, E\}$, for any $\mathcal{V} \subseteq V$, define

$$\partial(\mathcal{G}, \mathcal{V}) = \{j : i \in \mathcal{V}, (i, j) \in E\}.$$

Intuitively speaking, $\partial(\mathcal{G}, \mathcal{V})$ is the set of nodes in which each one has a link pointed from a corresponding node in $\mathcal{V}$.

A sequence of graphs $\{\mathcal{G}_i\}_{i=1}^m$ is jointly connected if $\bigcup_{i=1}^m \mathcal{G}_i$ contains a spanning tree (Jadbabaie et al., 2003).

A sequence of graphs $\{\mathcal{G}_i\}_{i=1}^m$ is sequentially connected (Angeli & Bliman, 2009) if there exists $\mathcal{V}_k \subseteq V$ ($1 \leq k \leq m+1$) satisfying²

$$\mathcal{V}_{k+1} \subseteq \partial(\mathcal{G}_k, \mathcal{S}_k), \quad \mathcal{S}_k = \bigcup_{s=1}^k \mathcal{V}_s,$$

where $\mathcal{S}_1 = \mathcal{V}_1$ is a singleton and $\mathcal{S}_{m+1} = V$.

The intuitive meaning of sequential connectivity can be explained as follows. When $\{\mathcal{G}_i\}_{i=1}^m$ is sequentially connected, there exists a node, called i^* (i.e. $\mathcal{S}_1 = \{i^*\}$), sends information to its neighbors $\mathcal{V}_2$ at time t_1 , then those who have received the information, called $\mathcal{S}_2$, send the information to their neighbors $\mathcal{V}_3$ at time t_2 , repeat the above process and finally all the nodes can get this information at time t_{m+1} .

3. Model, assumptions and main results

Let $V = \{1, 2, \dots, N\}$ be the set of nodes and consider the following complex network with nonlinear node dynamics (Yu et al., 2009)

$$\dot{x}_i(t) = f(x_i(t)) + c \sum_{j=1}^N a_{ij}(t)(x_j(t) - x_i(t)), \quad i \in V. \quad (1)$$

Here, $x_i(t) \in \mathbb{R}^n$, $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the nonlinear vector function, $c \in \mathbb{R}$ is the coupling coefficient, and $a_{ij}(t) \in \mathbb{R}$ are piecewise functions satisfying $a_{ij}(t) \equiv 0$ or $a_{ij}(t) \equiv 1$ in each interval $[t_k, t_{k+1})$,³ where $t_k = (k-1)h$ and h is the sampling step.⁴ Let $a_{ii}(t) = 0$ for any $t \geq 0$ and $i \in V$. Denote that

$$A_k = (a_{ij}(t))_{i,j=1}^N$$

in $[t_k, t_{k+1})$. Synchronization of complex networks (1) implies that

$$\lim_{t \rightarrow \infty} \|x_i(t) - x_j(t)\| = 0$$

for any $i, j \in V$.

The following assumptions are needed throughout the paper:

Assumption 1. The nonlinear function f is one-sided Lipschitz, which means there exists $L \in \mathbb{R}$ such that for any $x, y \in \mathbb{R}^n$, the following inequality holds:

$$(x - y)^T (f(x) - f(y)) \leq L \|x - y\|^2.$$

Assumption 2. There exists some integer $T > 0$ such that the sequence of graphs $\{\mathcal{G}_k\}_{k=(r-1)T+1}^r$ is sequentially connected for any $r \geq 1$, $ch \leq 1$, and

$$e^{LTh} (1 - (che^{-2(N-1)ch})^T) < 1. \quad (2)$$

Assumption 1 stresses the requirement for the nonlinear function $f(\cdot)$, **Assumption 2** establishes the relationship between coupling coefficient c and sampling interval h under the condition of sequentially connected topology, respectively.

According to Abbaszadeh and Marquez (2010), any Lipschitz function is one-sided Lipschitz, but the converse is not true. This means that for some nonlinear function with big Lipschitz constant, the corresponding one-sided Lipschitz constant L may be very small or even negative (see, the example given in Section 5). In the case that $L \leq 0$, it is easy to verify the condition 2 is naturally satisfied, leading to the unconservatism of the given assumptions.

The main result of this paper can be illustrated as follows.

Theorem 3. If **Assumptions 1** and **2** hold, then complex network (1) reaches synchronization.

³ We make this requirement only to simplify the technical proof, $a_{ij}(t)$ can be generalized to the time-varying case by using the same method of this paper.

⁴ By using the same method in Section 4, the sampling interval h can be generalized to non-evenly distributed h_k , we omitted the detailed deduction due to restriction of space.

² The definition of sequential connectivity for continuous-time networks (1) is different from that of discrete-time networks in that: there is no self-loop required in the continuous-time case.

According to [Theorem 3](#) and the definition of sequential connectivity, complex network (1) reaches synchronization even if the topology is not connected at any time instant.

The above theorem can be generalized to the case of joint connectivity by applying the following proposition.

Proposition 4 (Cao et al., 2008). Consider a sequence of graphs $\{\mathcal{G}_k\}_{k=1}^m$ with each $\mathcal{G}_k$ being connected and containing N nodes, then $\{\mathcal{G}_k\}_{k=1}^m$ is sequentially connected when $m \geq (N-1)^2$.

Since a sequence of connected graphs $\{\mathcal{G}_k\}_{k=1}^m$ can be viewed as sequentially connected if $m \geq (N-1)^2$. Consequently, if $\{\mathcal{G}_k\}_{k=(s-1)T+1}^{sT}$ is jointly connected for any $s \geq 1$, then $\{\mathcal{G}_k\}_{k=1}^{(N-1)^2T}$ is sequentially connected. In this sense, [Theorem 3](#) can be applied to the case of jointly connected graphs. The following result illustrates the synchronization conditions for jointly connected topology, whose proof will not be given in the paper since it can be derived directly from [Theorem 3](#).

Assumption 5. There exists some integer $T > 0$ such that the sequence of graphs $\{\mathcal{G}_k\}_{k=(r-1)T+1}^{rT}$ is jointly connected for any $r \geq 1$, $ch \leq 1$, and

$$e^{LKh} (1 - (che^{-2(N-1)ch})^K) < 1,$$

where $K = T(N-1)^2$.

Theorem 6. If [Assumptions 1](#) and [5](#) hold, then complex network (1) reaches synchronization.

The counterpart of network synchronization is the consensus of multi-agent systems, which has been widely investigated in related works, such as (Hong et al., 2007; Ren & Beard, 2005), and (Kim et al., 2013). Especially, the work of Kim et al. investigated the consensus of higher-order multi-agent systems with switching topology via an averaging approach, which is a typical method for the analysis of continuous-time switching systems (Kim et al., 2013). However, such an averaging method cannot be applied to the synchronization problem in this paper due to the following reasons: (a) the averaging method requires the averaged topology has some fixed structure, our paper only requires the topology to be sequentially connected, which means the averaged topology has no need to be a fixed one; (b) analysis via the averaging method needs the assistance of some specific linear transformations, but linear transformations are not applicable in our nonlinear synchronization model.

If $f(x) \equiv 0$, then (2) holds naturally. In this case, [Theorem 3](#) covers the results on consensus of multi-agent systems with switching topologies (Olfati-Saber & Murray, 2004).

If we define $g(h) = e^{LTh} (1 - (che^{-2(N-1)ch})^T)$ and let $ch = \frac{1}{2(N-1)}$, one obtains

$$g(h) = e^{\frac{LT}{2(N-1)}} \left(1 - \frac{1}{(2e(N-1))^T} \right)$$

and $\lim_{c \rightarrow \infty} g(h) < 1$, this implies a sufficient large coupling coefficient c synchronizes network (1).

4. Technical analysis

Define

$$e_{ij} = x_i - x_j, \quad F_{ij} = f(x_i) - f(x_j).$$

Note that

$$\begin{aligned} x_k - x_i &= x_k - x_j - (x_i - x_j) = e_{kj} - e_{ij}, \\ x_k - x_j &= x_k - x_i - (x_j - x_i) = e_{ki} - e_{ji}, \end{aligned}$$

one gets

$$\dot{e}_{ij} = F_{ij} + c \sum_{k=1, k \neq i}^N a_{ik}(t)(e_{kj} - e_{ij}) + c \sum_{k=1, k \neq j}^N a_{jk}(t)(e_{ik} - e_{ij}). \quad (3)$$

If [Assumption 2](#) is satisfied, there exists $\mathcal{V}_k \subseteq V$ ($1 \leq k \leq T+1$) satisfying

$$\mathcal{V}_{k+1} = \partial(\mathcal{G}_k, \mathcal{S}_k), \quad \mathcal{S}_k = \bigcup_{s=1}^k \mathcal{V}_s,$$

where $\mathcal{S}_1 = \mathcal{V}_1$ is a singleton, $\mathcal{S}_{T+1} = V$. The sequence of sets $\{\mathcal{V}_k\}_{k=1}^{T+1}$ is called a propagation sequence. Without loss of generality, let $\mathcal{V}_1 = \{1\}$. It is obvious that $\mathcal{S}_{k+1} = \mathcal{S}_k \cup \mathcal{V}_{k+1}$.

Denote

$$d(t) = \max_{i \neq j} \|e_{ij}(t)\|, \quad d = d(0).$$

Furthermore, for any $k \geq 1$, define

$$\hat{d}_k(t) = \max_{i, j \in \mathcal{S}_k} \|e_{ij}(t)\|, \quad t \in [t_k, t_{k+1}),$$

$$d_k = \max_{i, j \in \mathcal{S}_k} \|e_{ij}(t_k)\|.$$

In the following discussions, several lemmas are given to facilitate the proof of the main theorems. The proofs of the first four lemmas are omitted since they can be easily verified.

By using the following lemmas, one can obtain the proof of the main theorems step by step. The basic idea can be summarized as follows: the synchronization of network can be restated as $\lim_{t \rightarrow \infty} d(t) = 0$, which can be transformed to $\lim_{k \rightarrow \infty} d(t_k) = 0$, hence one only needs to show that the sequence d_k has a contractive property. To achieve such a destination, the central step is to build the connection between d_k and d_{k+1} by using the specific property of sequentially connected topology (see, [Lemma 11](#)), and then use iteration to get the relationship between d_k and d (see, [Lemma 12](#)). If there exists a contractive property between d_k and d , then network synchronization can be achieved.

Lemma 7. The solution for the differential equation $\dot{x} = \alpha x + \beta e^{Lt} + \gamma e^{\alpha t}$ with initial state $x(t_0) = x_0$ and $L \neq \alpha$ is given by $x(t) = x_0 e^{\alpha(t-t_0)} + \gamma e^{\alpha t} (t - t_0) + \frac{\beta}{L-\alpha} e^{Lt_0} (e^{L(t-t_0)} - e^{\alpha(t-t_0)})$.

Lemma 8. The solution for the discrete-time iteration $x_{k+1} = \alpha x_k + c\beta^k$ with $\alpha \neq \beta$ is given by $x_k = \frac{c\beta}{\beta-\alpha} (\beta^{k-1} - \alpha^{k-1}) + x_1 \alpha^{k-1}$.

Lemma 9. For $x(t) : \mathbb{R} \rightarrow \mathbb{R}^n$, it holds that $D^+(\|x(t)\|) \leq \|D^+(x(t))\|$, where $D^+(x(t)) = \limsup_{h \rightarrow 0^+} \frac{x(t+h) - x(t)}{h}$.

Lemma 10. Given $a, b \in \mathbb{R}^N$, if $\|a\| \geq \|b\|$, then $a^T(b - a) \leq 0$.

Lemma 11. For any $t \geq 0$, there is $d(t) \leq e^{Lt} d(0)$.

Proof. For any $t \geq 0$, there exist some $i, j \in V$ such that $\|e_{ij}(t)\| = d(t)$. For this i, j and t , one has

$$\begin{aligned} e_{ij}^T \dot{e}_{ij} &= e_{ij}^T F_{ij} + c e_{ij}^T \sum_{k=1, k \neq i}^N a_{ik}(t)(e_{kj} - e_{ij}) \\ &\quad + c e_{ij}^T \sum_{k=1, k \neq j}^N a_{jk}(t)(e_{ik} - e_{ij}). \end{aligned}$$

Since $\|e_{ij}\| \geq \|e_{kj}\|$ and $\|e_{ij}\| \geq \|e_{ik}\|$, according to [Lemma 10](#), one has $e_{ij}^T(e_{kj} - e_{ij}) \leq 0$ and $e_{ij}^T(e_{ik} - e_{ij}) \leq 0$. Therefore, $e_{ij}^T \dot{e}_{ij} \leq e_{ij}^T F_{ij} \leq L\|e_{ij}\|^2$ and

$$D^+(\|d(t)\|^2) = D^+(e_{ij}^T \dot{e}_{ij}) \leq 2L\|e_{ij}\|^2 = 2L\|d(t)\|^2.$$

Consequently, $d(t) \leq e^{Lt} d(0)$ and the proof is completed. ■

Lemma 12. If [Assumptions 1](#) and [2](#) hold, then for $t \in [t_k, t_{k+1})$, it holds that

$$\hat{d}_k(t) \leq d_k e^{\lambda(t-t_k)} + d e^{L t_k} (e^{L(t-t_k)} - e^{\lambda(t-t_k)}),$$

where $\lambda = L - 2c(N - 1)$.

Proof. Denote closed unit ball $\mathbb{B} = \{x \in \mathbb{R}^n : \|x\| \leq 1\}$. According to [Lemma 11](#), one has $e_{ik}(t), e_{kj}(t) \in e^{L t} d \cdot \mathbb{B}$ for any $i, j, k \in V$. Furthermore, for any $i, j \in \mathcal{S}_k$, and $t \in [t_k, t_{k+1})$, using of [\(3\)](#) leads to

$$\dot{e}_{ij} \in F_{ij} + c \left(\sum_{k=1, k \neq i}^N a_{ik}(t) + \sum_{k=1, k \neq j}^N a_{jk}(t) \right) (e^{L t} d \cdot \mathbb{B} - e_{ij}).$$

If $e_{ij} \neq 0$, one has

$$e_{ij}^T \dot{e}_{ij} \in e_{ij}^T F_{ij} + c \left(\sum_{k=1, k \neq i}^N a_{ik}(t) + \sum_{k=1, k \neq j}^N a_{jk}(t) \right) (e^{L t} d \cdot e_{ij}^T \mathbb{B} - e_{ij}^T e_{ij}).$$

Based on the facts $e_{ij}^T \dot{e}_{ij} = \frac{1}{2} D^+(e_{ij}^T e_{ij}) = \frac{1}{2} D^+(\|e_{ij}\|^2) = \|e_{ij}\| \cdot D^+(\|e_{ij}\|)$ and $e_{ij}^T F_{ij} \leq L \|e_{ij}\|^2$, $\sup e_{ij}^T \mathbb{B} \leq \|e_{ij}\|$, $e_{ij}^T e_{ij} = \|e_{ij}\|^2$, one gets

$$\begin{aligned} D^+(\|e_{ij}\|) &\leq L \|e_{ij}\| + c \left(\sum_{k=1, k \neq i}^N a_{ik}(t) + \sum_{k=1, k \neq j}^N a_{jk}(t) \right) (e^{L t} d - \|e_{ij}\|) \\ &\leq L \|e_{ij}\| + 2c(N - 1)(e^{L t} d - \|e_{ij}\|) \\ &= (L - 2c(N - 1))\|e_{ij}\| + 2c(N - 1)d \cdot e^{L t}. \end{aligned} \quad (4)$$

If $e_{ij} = 0$, then $x_i = x_j$ and $F_{ij} = 0$. One derives $\dot{e}_{ij} \in c \left(\sum_{k=1, k \neq i}^N a_{ik}(t) + \sum_{k=1, k \neq j}^N a_{jk}(t) \right) (e^{L t} d \cdot \mathbb{B})$. Hence, $\|\dot{e}_{ij}\| \leq c \left(\sum_{k=1, k \neq i}^N a_{ik}(t) + \sum_{k=1, k \neq j}^N a_{jk}(t) \right) e^{L t} d$. According to [Lemma 9](#), one has

$$\begin{aligned} D^+(\|e_{ij}\|) &\leq c \left(\sum_{k=1, k \neq i}^N a_{ik}(t) + \sum_{k=1, k \neq j}^N a_{jk}(t) \right) e^{L t} d \\ &\leq 2c(N - 1)d \cdot e^{L t}. \end{aligned} \quad (5)$$

Summarize [\(4\)](#), [\(5\)](#) and according to [Lemma 7](#), one gets

$$\begin{aligned} \|e_{ij}(t)\| &\leq e^{(L-2c(N-1))(t-t_k)} \|e_{ij}(t_k)\| \\ &\quad + d e^{L t_k} (e^{L(t-t_k)} - e^{(L-2c(N-1))(t-t_k)}). \end{aligned}$$

Therefore, $\hat{d}_k(t) \leq d_k e^{\lambda(t-t_k)} + d e^{L t_k} (e^{L(t-t_k)} - e^{\lambda(t-t_k)})$ and the proof is completed. ■

Lemma 13. If [Assumptions 1](#) and [2](#) hold, then for any $k \geq 1$, it holds that

$$d_{k+1} \leq (e^{L h} - \mu^k) d,$$

where $\mu = che^{\lambda h}$ and $\lambda = L - 2(N - 1)c$.

Proof. For any $i \in \mathcal{S}_{k+1}$ and $j \in \mathcal{S}_{k+1}$, since $\mathcal{S}_{k+1} = \mathcal{S}_k \cup \mathcal{V}_{k+1}$, one can derive the proof of this lemma from the following three cases.

- (a) $i \in \mathcal{S}_k$ and $j \in \mathcal{V}_{k+1}$;
- (b) $i, j \in \mathcal{S}_k$;
- (c) $i, j \in \mathcal{V}_{k+1}$.

CASE (a): For $i \in \mathcal{S}_k$ and $j \in \mathcal{V}_{k+1}$, there exists $r \in \mathcal{S}_k$ such that $(r, j) \in \mathcal{G}_k$. Hence, $a_{jr}(t) = 1$ for $t \in [t_k, t_{k+1})$. Based on [\(3\)](#), one

has

$$\begin{aligned} \dot{e}_{ij} &= F_{ij} + c(e_{ir} - e_{ij}) + c \sum_{k=1, k \neq i}^N a_{ik}(t)(e_{kj} - e_{ij}) \\ &\quad + c \sum_{k=1, k \neq j, r}^N a_{jk}(t)(e_{ik} - e_{ij}) \\ &\in F_{ij} + c \cdot (\hat{d}_k(t) \mathbb{B} - e_{ij}) + c \left(\sum_{k=1, k \neq i}^N a_{ik}(t) \right. \\ &\quad \left. + \sum_{k=1, k \neq j, r}^N a_{jk}(t) \right) (e^{L t} d \cdot \mathbb{B} - e_{ij}). \end{aligned}$$

Similar to the deduction of [\(4\)](#) and [\(5\)](#), there is

$$\begin{aligned} D^+(\|e_{ij}\|) &\leq L \|e_{ij}\| + c(\hat{d}_k(t) - \|e_{ij}\|) \\ &\quad + c \left(\sum_{k=1, k \neq i}^N a_{ik}(t) + \sum_{k=1, k \neq j, r}^N a_{jk}(t) \right) (e^{L t} d - \|e_{ij}\|) \\ &\leq L \|e_{ij}\| + c(\hat{d}_k(t) - \|e_{ij}\|) + c(2N - 3)(e^{L t} d - \|e_{ij}\|) \\ &= (L - 2(N - 1)c)\|e_{ij}\| + c(2N - 3)d e^{L t} + c \cdot \hat{d}_k(t) \\ &\leq (L - 2(N - 1)c)\|e_{ij}\| + c(2N - 3)d e^{L t} \\ &\quad + c(d_k e^{\lambda(t-t_k)} + d e^{L t} - d e^{\lambda(t-t_k)} e^{L t_k}) \\ &= (L - 2(N - 1)c)\|e_{ij}\| + c(2N - 2)d e^{L t} \\ &\quad + c e^{\lambda(t-t_k)} (d_k - d e^{L t_k}). \end{aligned}$$

According to [Lemma 7](#), one has

$$\begin{aligned} \|e_{ij}(t)\| &\leq \|e_{ij}(t_k)\| e^{\lambda(t-t_k)} + d e^{L t_k} (e^{L(t-t_k)} - e^{\lambda(t-t_k)}) \\ &\quad + c(t - t_k) e^{\lambda(t-t_k)} (d_k - d e^{L t_k}). \end{aligned}$$

Consequently, one gets

$$\begin{aligned} \|e_{ij}(t_{k+1})\| &\leq \|e_{ij}(t_k)\| e^{\lambda h} + d e^{L t_k} (e^{L h} - e^{\lambda h}) + c h e^{\lambda h} (d_k - d e^{L t_k}) \\ &\leq d(e^{L t_k} e^{\lambda h} + e^{L t_k} (e^{L h} - e^{\lambda h}) - c h e^{\lambda h} e^{L t_k}) + d_k c h e^{\lambda h} \\ &= d e^{L t_k} (e^{L h} - c h e^{\lambda h}) + d_k c h e^{\lambda h}. \end{aligned}$$

CASE (b): For $i, j \in \mathcal{S}_k$, according to [Lemma 12](#) and $ch \leq 1$, there is

$$\begin{aligned} \|e_{ij}(t_{k+1})\| &\leq \hat{d}_k(t_{k+1}) \\ &\leq d_k e^{\lambda h} + d e^{L t_k} (e^{L h} - e^{\lambda h}) \\ &\leq d e^{L t_k} (e^{L h} - c h e^{\lambda h}) + d_k c h e^{\lambda h}. \end{aligned}$$

CASE (c): For $i, j \in \mathcal{V}_{k+1}$, there exist $r_1, r_2 \in \mathcal{S}_k$ such that $(r_1, i), (r_2, j) \in \mathcal{G}_k$. Hence, for $t \in [t_k, t_{k+1})$, one has $a_{ir_1}(t) = a_{jr_2}(t) = 1$. Consequently,

$$\begin{aligned} \dot{e}_{ij} &= F_{ij} + c(e_{r_1 i} - e_{ij}) + c(e_{r_2 j} - e_{ij}) \\ &\quad + c \sum_{k=1, k \neq i, r_1}^N a_{ik}(t)(e_{kj} - e_{ij}) + c \sum_{k=1, k \neq j, r_2}^N a_{jk}(t)(e_{ik} - e_{ij}). \end{aligned}$$

From the definition of e_{ij} , based on the fact of $e_{r_1 i} + e_{r_2 j} = e_{ij} + e_{r_1 r_2}$, one has

$$\begin{aligned} \dot{e}_{ij} &= F_{ij} + c(e_{r_1 r_2} - e_{ij}) + c \sum_{k=1, k \neq i, r_1}^N a_{ik}(t)(e_{kj} - e_{ij}) \\ &\quad + c \sum_{k=1, k \neq j, r_2}^N a_{jk}(t)(e_{ik} - e_{ij}) \end{aligned}$$

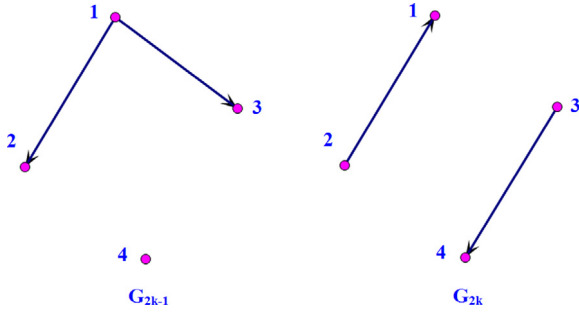


Fig. 1. Switching topology at $t = t_{2k-1}$ and $t = t_{2k}$ for $k \geq 1$.

$$\in F_{ij} + c \cdot (\hat{d}_k(t) \mathbb{B} - e_{ij}) + c \left(\sum_{k=1, k \neq i, r_1}^N a_{ik}(t) + \sum_{k=1, k \neq j, r_2}^N a_{jk}(t) \right) (e^{Lt} d \cdot \mathbb{B} - e_{ij}).$$

Similar to the proof of CASE (a), one can also derive that

$$\|e_{ij}(t_{k+1})\| \leq d e^{L t_k} (e^{L h} - c h e^{\lambda h}) + d_k c h e^{\lambda h}.$$

Summarize the results of the above three cases and according to Lemma 8, one has

$$d_{k+1} \leq d(e^{L(k-1)h} - \mu^{k-1}) + d_2 \mu^{k-1}, \quad \mu = c h e^{\lambda h}.$$

Note that the discussion on CASE (a)–(c) is also applicable to the case of $k = 1$. Considering $d_1 = 0$, $t_1 = 0$, one gets

$$d_2 \leq d e^{L t_1} (e^{L h} - c h e^{\lambda h}) + d_1 c h e^{\lambda h} = d(e^{L h} - c h e^{\lambda h}).$$

Since $\mu = c h e^{\lambda h}$, one knows that $\mu < e^{L h}$. Therefore,

$$\begin{aligned} d_{k+1} &\leq d(e^{L(k-1)h} - \mu^{k-1}) + d(e^{L h} - c h e^{\lambda h}) \mu^{k-1}, \\ &= d((e^{L h})^{k-1} + (e^{L h} - 1) \mu^{k-1} - \mu^k) \\ &\leq d((e^{L h})^{k-1} + (e^{L h} - 1) e^{L h(k-1)} - \mu^k) \\ &= d(e^{L k h} - \mu^k). \end{aligned}$$

This completes the proof. ■

Based on the above lemmas, one gives the proof of Theorem 3 as follows.

Proof of Theorem 3. According to Lemma 13, one has

$$d_{T+1} \leq (e^{L T h} - \mu^T) d = e^{L T h} (1 - (c h e^{-2(N-1)ch})^T) d.$$

Similarly, $d_{sT+1} \leq e^{L T h} (1 - (c h e^{-2(N-1)ch})^T) d_{(s-1)T+1}$ for any $s \geq 1$. This completes the proof. ■

5. An illustrative example

Consider the nonlinear dynamics

$$f(z) = \begin{pmatrix} z_1 - z_2 - z_1(z_1^2 + z_2^2) \\ z_1 + z_2 - z_2(z_1^2 + z_2^2) \end{pmatrix}$$

with $z = (z_1, z_2)^T \in \mathbb{R}^2$. As pointed out in Abbaszadeh and Marquez (2010), the one-sided Lipschitz constant for the above $f(\cdot)$ is $L = 0$. However, the Lipschitz constant of $f(\cdot)$ in the region $\mathcal{D} = \{z \in \mathbb{R}^2 : \|z\| \leq r\}$ is $3r^2$, such a value exceeds 300 once $r \geq 10$. This example demonstrates the advantage of one-sided Lipschitz constant in characterizing the complexity of nonlinear dynamics.

Given complex networks (1) with $N = 4$ nodes and $x_i = (x_i^{(1)}, x_i^{(2)})^T \in \mathbb{R}^2$. Set $h = 0.1$ and $c = 5$ and the switching topology is given in Fig. 1.

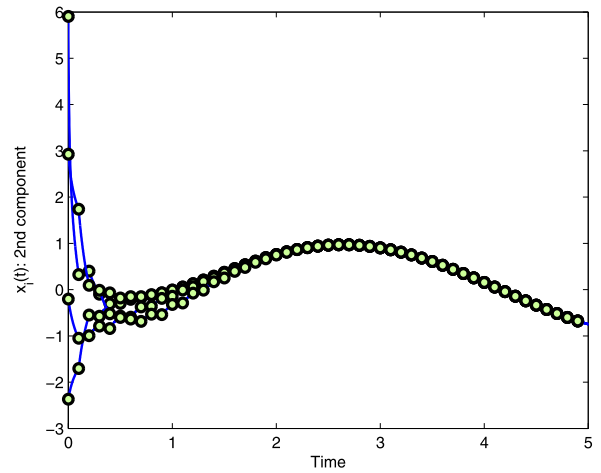
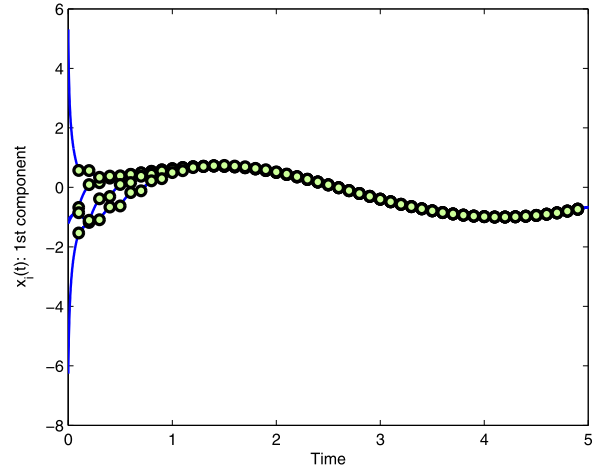


Fig. 2. Trajectories of the 4 nodes with a random set of initial states.

Consider the propagation sequence $\{1\} \rightarrow \{2, 3\} \rightarrow \{4\}$, it is easy to verify that $\{g_{2k-1}, g_{2k}\}$ is sequentially connected (also jointly connected), where the constant T in Assumption 2 is $T = 2$. One can easily verify that both Assumptions 1 and 2 are satisfied.

Let the initial states of the network be randomly chosen from $[-10, 10] \times [-10, 10]$. The set of trajectories is given in Fig. 2. It should be noted that the circles in Fig. 2 represent the switching times.

6. Concluding remarks

By employing the concepts of joint connectivity and sequential connectivity, this paper reports a novel result on synchronization of complex networks: nonlinear complex networks can reach synchronization even if the network topology is not connected at any time instant. The intuitive explanation for such a phenomenon can be given as follows: sequentially connected topology generates a contractive force to bring the network states together, if this contractive force is greater than the divergence force brought by the node nonlinearity, then the network synchronizes. The results of this paper provide insights into the synchronization mechanism of complex networks and will possibly have potential applications in real engineering systems.

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Yao Chen received the B.S. degree in mathematics from Three Gorges University, Yichang, China, in 2007, and the Ph.D. degree in control theory from the Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing, China, in 2012.

He is currently a Postdoc Research Fellow with the Department of Computing, The Hong Kong Polytechnic University, Hong Kong. His research interests include complex networks, multiagent systems, and rail traffic control and optimization.



Wenwu Yu received the B.Sc. degree in information and computing science and M.Sc. degree in applied mathematics from the Department of Mathematics, Southeast University, Nanjing, China, in 2004 and 2007, respectively, and the Ph.D. degree from the Department of Electronic Engineering, City University of Hong Kong, Hong Kong, China, in 2010. Currently, he is the Founding Director of Laboratory of Cooperative Control of Complex Systems, an Associate Director in the Research Center for Complex Systems and Network Sciences, an Associate Dean in the Department of Mathematics, and a Full Professor with the Distinguished Young Honor in Southeast University, China.

He has authored or coauthored about 90 refereed international journal papers, and is a reviewer of several journals. His research interests include multi-agent systems, complex networks and systems, nonlinear dynamics and control, smart grids, neural networks, game theory, robotics, and optimization. He was listed by Thomson Reuters Highly Cited Researchers in Engineering in 2014 and 2015, respectively.



Shaolin Tan received the B.E. degree in Electrical Science and Technology from the Zhengzhou University, Zhengzhou, China in 2009, and the Ph.D. degree from the Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing, China in 2014.

Currently, he is an assistant professor with the College of Electrical and Information Engineering, Hunan University. His research interests include complex networks, evolutionary game theory, multi-agent systems, and swarm intelligence.



Henghui Zhu received the M.Sc. degree in systems theory from the Academy of Mathematics and Systems Science, Chinese Academy of Sciences in 2015 and B.Sc. degree in mathematics from Wuhan University in 2012. Currently, he is working towards the Ph.D. degree in Division of Systems Engineering, Boston University. His research interests include optimization, machine learning and multi-agent systems.